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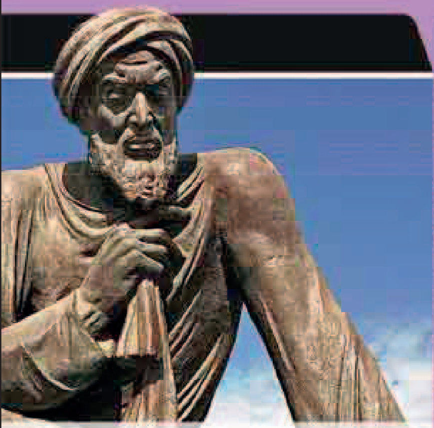
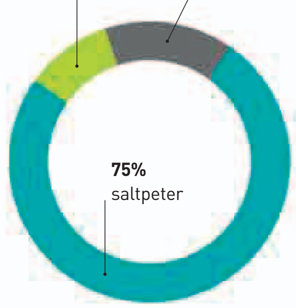


This statue of Al-Khwarizmi stands in Khiva, Uzbekistan, his birthplace.

The Chinese edition of the Buddhist *Diamond Sutra*, printed using woodblocks on a scroll of paper, is the earliest surviving printed book.

transmutation of metals. However, alchemy was at the root of another discovery—this time in China. In the early 9th century, Chinese alchemists were experimenting with various mixtures of substances to find the "elixir" of life. One of the by-products of this quest was the **discovery**, in about 855, of **gunpowder**—the first man-made explosive. It consisted of a mixture of sulfur, carbon (in the form of charcoal), and saltpeter (potassium nitrate) – all of which occur naturally as minerals. The mixture's explosive properties meant that it was initially used in the **manufacture of fireworks**, but gunpowder later came to fuel rockets, and was eventually used in the development of firearms.



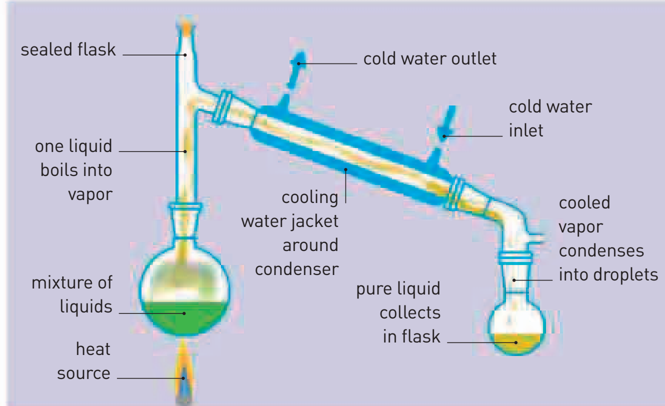
Composition of gunpowder
Sulfur, carbon, and saltpeter, while quite innocuous individually, become highly explosive when mixed in the correct proportions.

A CHINESE EDITION OF THE BUDDHIST TEXT, *Diamond Sutra*, was discovered in 1907 in Dunhuang, northwest China. Although it is probably not the first example of a **woodblock printed book**, it is the earliest known one, and bears the date May 11, 868. The text and illustrations of *Diamond Sutra* exhibit a great deal of sophistication, suggesting that the techniques of printing on paper were well known in China by this time. An inscription at the end of the manuscript indicates that this was one of a number of copies printed for distribution.

An inscription on a stone in Gwalior, India, dated 876, contains one of the earliest known uses of the **symbol for zero—"0"**. Prior to the appearance of a specific symbol, a space was used to indicate zero, which led to ambiguity and prevented the development of a place value system of numbers (a system in which the position of the numeral indicates its value). The introduction of a symbol for zero in Indian mathematics was a vital step in the development of the **decimal system of notation** we use today. This decimal system came to Europe through the influence of Islamic mathematicians, and eventually replaced the use of cumbersome Roman numerals.

Alchemist Jabir ibn-Hayyan at work
Alchemy in the Islamic world involved much experimentation, and led to the development of many processes that were later used in chemistry.

Toward the end of the century, Arab alchemists developed the process of **distillation**—a method of separating the ingredients of a liquid mixture. **Muhammad ibn Zakariya al-Razi** (c.854–925/35), along with other alchemists, perfected the technique and was successful in extracting a form of alcohol—ethanol or ethyl alcohol—by distilling wine. The word alcohol derives from the Arabic *al kuhl*, originally used to describe a powder extracted from a mineral, but which later came to mean the essence or "spirit" of a liquid. The apparatus developed by al-Razi for distillation has remained fundamentally unchanged to the present day.



DISTILLATION

Distillation is a method of separating the components of a liquid mixture. The liquid mixture is converted into vapor by heating. As the components of the mixture have different boiling points, they vaporize at different rates. The vapor is then cooled so that it condenses back into a liquid, which can be collected separately. Distillation can be used to extract liquids such as alcohol and gasoline, and also to purify liquids, such as salt water.

868 The earliest surviving printed book, the ***Diamond Sutra***, is printed in China

876 Indian mathematicians use a **symbol for zero**

c.890 Al-Razi distils alcohol from wine